Bellabeat Case Study

From Data to Decisions: A Deep Dive into User Behaviour

SQL Analysis | Tableau Visualisation | Strategic Insights

Introduction

Bellabeat's Mission:
Empowering women through wellness tech.

Purpose of the case study:
Show how data → insights → strategy.

Dataset Context

Source:

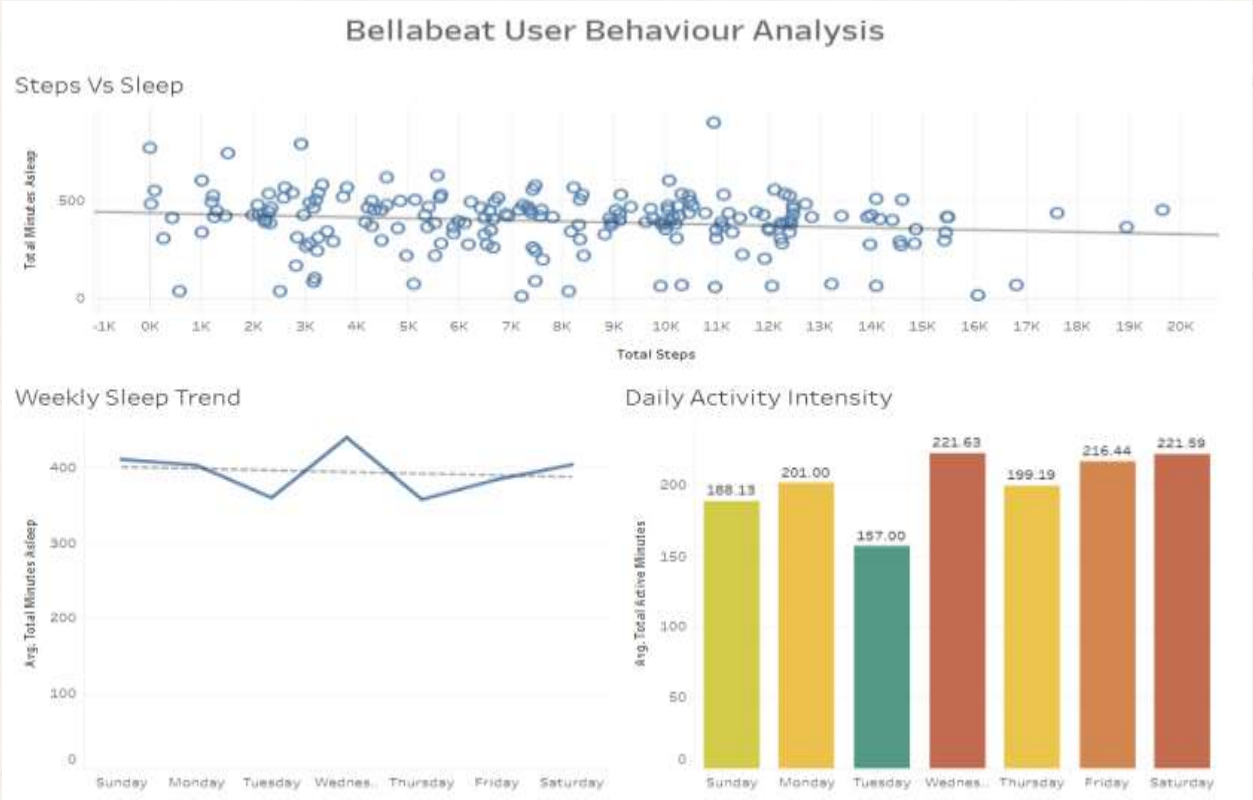
Bellabeat Fitness Tracker Dataset (via Kaggle).

Key Variables:

- Daily Steps
- Calories
- Active vs. Sedentary Minutes
- Sleep Duration

Primary Objective:

Analyze user engagement patterns to unlock growth opportunities for Bellabeat's smart products.



Key Takeaway

While daily activity is high, there is a visible gap in consistent sleep logging, suggesting users view the device primarily as a fitness tracker rather than a wellness tool.

Core Data Insights

Consistency

Activity tracking is high across the user base, indicating strong daily habit formation.

The Link

A near-linear correlation exists between step counts and calories burned, validating the accuracy of the tracking.

The Gap

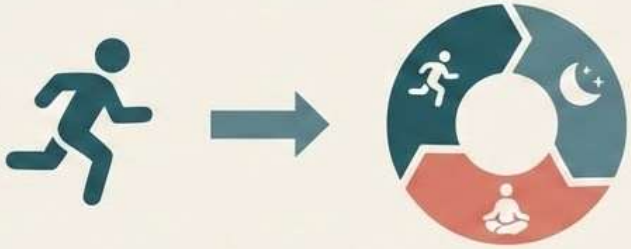
Sedentary minutes remain high even for “active” users—revealing a prime opportunity for real-time behavioral nudges.

The Holistic Wellness Gap

The Findings

Activity tracking significantly outweighs sleep tracking.

Strategic Takeaway



Current Focus: Fitness Leader

Future Opportunity: Holistic Health Leader

Users focus heavily on ‘grind’ metrics (movement) but overlook ‘recovery’ metrics (sleep). Bellabeat has the opportunity to **redefine** itself as the **leader in holistic health**, not just fitness.

Marketing Strategy & Recommendations

Personalized Nudges

Implement "Smart Sleep Reminders" that suggest optimal bedtimes based on the day's activity level.

Gamification

Introduce badges for "Balanced Days" where users meet both their step goals and sleep targets.

Campaign Concept

"Complete Your Wellness Circle" — A marketing push encouraging users to sync activity, sleep, and mindfulness for a 360-degree view of health.

Technical Methodology

BigQuery (SQL)

- Used for data cleaning, joining multiple datasets, and aggregating daily activity and sleep metrics.

Tableau

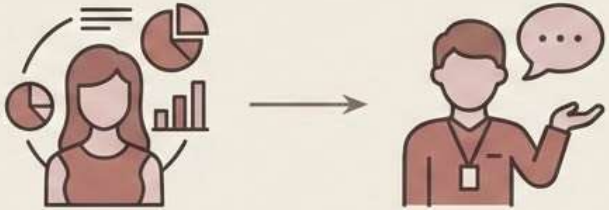
- Developed an interactive dashboard to visualize correlations between activity levels and sleep patterns.

Google Sheets

- Utilized for initial data exploration and formatting.

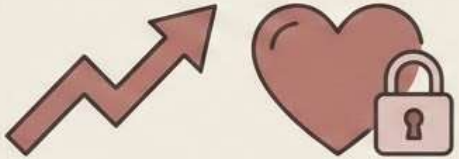
Conclusion

The Story



By leveraging these datasets, Bellabeat can transition from a data provider to a wellness coach.

Impact



Bridging the gap between activity and recovery will drive higher app engagement and long-term product loyalty.

Technical Appendix I: Data Standardization

```sql
WITH daily_sleep AS (
  SELECT
    Id,
    -- Normalizing inconsistent date formats for join accuracy
    PARSE_DATE('%m/%d/%Y', SPLIT(date, ' ')[OFFSET(0)]) AS sleep_date,
    COUNT(value) AS total_minutes_asleep
  FROM `nifty-yeti-492810-d4.bellabeat_data.minuteSleep_merged`
  WHERE value = 1
  GROUP BY Id, sleep_date
),

daily_weight AS (
  SELECT
    Id,
    PARSE_DATE('%m/%d/%Y', SPLIT(Date, ' ')[OFFSET(0)]) AS weight_date,
    WeightKg,
    BMI
  FROM `nifty-yeti-492810-d4.bellabeat_data.weightLogInfo_merged`
)

-- Final Selection & Multi-Table Join
SELECT
  A.Id,
  A.ActivityDate,
  A.TotalSteps,
  -- Aggregating all active tiers into a single 'Total Active' metric
  (A.VeryActiveMinutes + A.FairlyActiveMinutes + A.LightlyActiveMinutes) AS
toSatgaattmetsvmcentosleep,
  W.WeightKg,
  W.BMI
FROM `nifty-yeti-492810-d4.bellabaat_data.dailyActivity_merged` AS A
LEFT JOIN daily_sleep AS S
  ON A.ID = S.ID AND A.ActivityDate = S.sleep_date,
LEFT JOIN daily_weight AS W
  ON A.ID = W.Id AND A.ActivityDate = W.weight_date;
);
```

The Challenge:

Raw datasets contained inconsistent date/time strings (MM/DD/YYYY HH:MM:SS) which prevented table joins.

The Solution:

Used SPLIT and PARSE_DATE functions in BigQuery to isolate dates and convert them into standardized DATE objects.

Key Skill Highlight:

String manipulation and data type casting.

Technical Appendix II: Behavioral Segmentation

```sql
/*
 STEP: USER CATEGORIZATION
 Objective: Segment users by engagement level based on
the number of days they tracked activity.
*/

SELECT
  Id,
  COUNT(ActivityDate) AS days_tracked,
  CASE
    WHEN COUNT(ActivityDate) >= 25 THEN 'Active User'
    WHEN COUNT(ActivityDate) BETWEEN 15 AND 24 THEN 'Moderate User'
    ELSE 'Light User'
  END AS user_type
FROM `nifty-yeti-492810-d4.bellabeat_data.dailyActivity_merged`
GROUP BY Id
ORDER BY days_tracked DESC;
```

The Logic:

Categorized users into three distinct tiers based on monthly engagement: Active (25+ days), Moderate (15-24 days), and Light (<15 days).

Strategic Purpose:

This segmentation allows Bellabeat to target 'Light Users' with retention campaigns and 'Active Users' with premium features or advanced tracking tools.

Key Skill Highlight:

Advanced conditional logic using CASE statements and GROUP BY aggregations.

Technical Appendix III: Statistical Aggregation

```sql
/*
 STEP: STATISTICAL AGGREGATION
 Objective: Calculate high-level averages for sedentary vs.
high-intensity behavior across the user base.
*/

SELECT

  -- Converting minutes to hours for better readability
  AVG(SedentaryMinutes) / 60 AS avg_sedentary_hours_per_day,

  -- Combining activity tiers to define "High Intensity"
  AVG(VeryActiveMinutes + FairlyActiveMinutes) AS
avg_high_intensity_minutes_per_day

FROM `nifty-yoti-492810-d4.bellabeat_data.dailyActivity_merged`;
```

The Insight:

Calculated that the average user spends over 16 hours per day sedentary, compared to only ~30 minutes of high-intensity activity.

Data Transformation: Performed real-time unit conversion (minutes to hours) within the SQL query to ensure results were 'human-readable' for stakeholders.

Business Impact: This specific data point formed the foundation for the recommendation to implement 'Sedentary Nudges' in the Bellabeat app

Key Skill Highlight:

Mathematical operations and statistical averaging in BigQuery.